

Solution of Exercise 4

Exercise 4.1 Calculus of variations

The *Gâteaux derivative* of J at the optimal point $\mathbf{x}^*$ in the direction of $\delta \mathbf{x}$ calculates to

$$\begin{aligned}\delta J(\mathbf{x}^*; \delta \mathbf{x}) &= \left. \frac{dJ(\mathbf{x}^* + \epsilon \delta \mathbf{x})}{d\epsilon} \right|_{\epsilon=0} \\ &= \int_{t_0}^{t_f} \left. \frac{d}{d\epsilon} \phi(\mathbf{x}^*(t) + \epsilon \delta \mathbf{x}(t), \dot{\mathbf{x}}^*(t) + \epsilon \delta \dot{\mathbf{x}}(t), t) \right|_{\epsilon=0} dt \\ &= \int_{t_0}^{t_f} \phi_{\mathbf{x}}(\mathbf{x}^*(t), \dot{\mathbf{x}}^*(t))^T \delta \mathbf{x}(t) + \phi_{\dot{\mathbf{x}}}(\mathbf{x}^*(t), \dot{\mathbf{x}}^*(t))^T \delta \dot{\mathbf{x}}(t) dt. \end{aligned} \quad (1)$$

In the next step the second term in (1) is being *partially integrated*¹ (neglecting the time arguments)

$$\int_{t_0}^{t_f} \phi_{\dot{\mathbf{x}}}(\mathbf{x}^*, \dot{\mathbf{x}}^*, t)^T \delta \dot{\mathbf{x}} dt = \underbrace{\left[\phi_{\dot{\mathbf{x}}}(\mathbf{x}^*, \dot{\mathbf{x}}^*, t)^T \delta \mathbf{x} \right]_{t_0}^{t_f}}_{\stackrel{!}{=} 0} - \int_{t_0}^{t_f} \frac{d}{dt} \phi_{\dot{\mathbf{x}}}(\mathbf{x}^*, \dot{\mathbf{x}}^*, t)^T \delta \mathbf{x} dt. \quad (2)$$

The new *natural boundary conditions* follow as

$$\begin{aligned}\phi_{\dot{\mathbf{x}}}(\mathbf{x}^*(t_0), \dot{\mathbf{x}}^*(t_0), t_0) &= 0 \\ \phi_{\dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) &= 0\end{aligned}$$

and analogously to the derivation in the script

$$\delta J(\mathbf{x}^*; \delta \mathbf{x}) = \int_{t_0}^{t_f} \left[\phi_{\mathbf{x}}(\mathbf{x}^*, \dot{\mathbf{x}}^*, t) - \frac{d}{dt} \phi_{\dot{\mathbf{x}}}(\mathbf{x}^*, \dot{\mathbf{x}}^*, t) \right]^T \delta \mathbf{x} dt \quad (3)$$

and thus the *Euler-Lagrange equations*

$$\phi_{\mathbf{x}}(\mathbf{x}^*(t), \dot{\mathbf{x}}^*(t), t) - \frac{d}{dt} \phi_{\dot{\mathbf{x}}}(\mathbf{x}^*(t), \dot{\mathbf{x}}^*(t), t) = \mathbf{0}. \quad (4)$$

¹Reminder: $\int \mathbf{u}^T \dot{\mathbf{v}} dt = \mathbf{u}^T \mathbf{v} - \int \dot{\mathbf{u}}^T \mathbf{v} dt$

Exercise 4.2 Dynamic optimization with free terminal state

- Hamiltonian function

$$\begin{aligned} H(x, u, \lambda) &= l(x, u) + \lambda f(x, u) \\ &= x^2 + u^2 + \lambda(u - x) \end{aligned}$$

- Optimality conditions

$$\dot{x} = u - x, \quad x(0) = 1 \quad (5)$$

$$\dot{\lambda} = -\frac{\partial H(x, u, \lambda)}{\partial x} = -2x + \lambda, \quad \lambda(1) = 0 \quad (6)$$

$$0 = \frac{\partial H(x, u, \lambda)}{\partial u} = 2u + \lambda \quad \Rightarrow \quad u = -\frac{\lambda}{2} \quad (7)$$

- Inserting the solution of the canonical equations and checking for equality

$$\begin{aligned} x^*(t) &= C \exp(-\sqrt{2}t) + D \exp(\sqrt{2}t) \\ u^*(t) &= C(1 - \sqrt{2}) \exp(-\sqrt{2}t) + D(1 + \sqrt{2}) \exp(\sqrt{2}t) \end{aligned}$$

Verification for x :

$$\dot{x}^* = -\sqrt{2}C \exp(-\sqrt{2}t) + \sqrt{2}D \exp(\sqrt{2}t) = u^* - x^* \quad \checkmark$$

Verification for λ : ($\dot{\lambda}^* \stackrel{!}{=} -2x^* + \lambda^* = -2x^* - 2u^*$)

$$\lambda^* = -2u^* = -2[C(1 - \sqrt{2}) \exp(-\sqrt{2}t) + D(1 + \sqrt{2}) \exp(\sqrt{2}t)]$$

$$\begin{aligned} \dot{\lambda}^* &= -2[-\sqrt{2}C(1 - \sqrt{2}) \exp(-\sqrt{2}t) + \sqrt{2}D(1 + \sqrt{2}) \exp(\sqrt{2}t)] \\ &= -2[C(2 - \sqrt{2}) \exp(-\sqrt{2}t) + D(2 + \sqrt{2}) \exp(\sqrt{2}t)] \\ &= -2[C(1 + 1 - \sqrt{2}) \exp(-\sqrt{2}t) + D(1 + 1 + \sqrt{2}) \exp(\sqrt{2}t)] \\ &= -2[x^* + u^*] \quad \checkmark \end{aligned}$$

Verification of the constants:

$$x(0) = 1 \quad \Rightarrow \quad C + D = 1 \quad \Rightarrow \quad C = 1 - D \quad (8)$$

$$\lambda(1) = 0 \quad \Rightarrow \quad -2[C(1 - \sqrt{2}) \exp(-\sqrt{2}) + D(1 + \sqrt{2}) \exp(\sqrt{2})] = 0 \quad (9)$$

$$\begin{aligned} &\Rightarrow (1 - D)(-2(1 - \sqrt{2}) \exp(-\sqrt{2})) + D(-2(1 + \sqrt{2}) \exp(\sqrt{2})) = 0 \\ D[(+2(1 - \sqrt{2}) \exp(-\sqrt{2})) + (-2(1 + \sqrt{2}) \exp(\sqrt{2}))] + (-2(1 - \sqrt{2}) \exp(-\sqrt{2})) &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow D &= \frac{-(-2(1 - \sqrt{2}) \exp(-\sqrt{2}))}{(+2(1 - \sqrt{2}) \exp(-\sqrt{2})) + (-2(1 + \sqrt{2}) \exp(\sqrt{2}))} \\ &= \frac{(-1 + \sqrt{2}) \exp(-\sqrt{2})}{(-1 + \sqrt{2}) \exp(-\sqrt{2}) + (1 + \sqrt{2}) \exp(\sqrt{2})} \end{aligned}$$

with $\sinh(x) = \frac{e^x - e^{-x}}{2}$ and $\cosh(x) = \frac{e^x + e^{-x}}{2}$ follows

$$D = \frac{(\sqrt{2} - 1) \exp(-\sqrt{2})}{2(\sinh(\sqrt{2}) + \sqrt{2} \cosh(\sqrt{2}))}$$

$$\text{und } C = 1 - D = \dots = \frac{(1 + \sqrt{2}) \exp(\sqrt{2})}{2(\sinh(\sqrt{2}) + \sqrt{2} \cosh(\sqrt{2}))}$$

Exercise 4.3 Dynamic optimization with fixed terminal state

- a) • Hamiltonian function $H(x, \lambda, u) = \frac{a}{2}u^2 + \frac{1}{2}x^2 + \lambda u$
 • Optimality conditions

$$\begin{aligned}\dot{x} &= \frac{\partial H}{\partial \lambda} = u & x(0) &= 1, \quad x(1) = 0 \\ \dot{\lambda} &= -\frac{\partial H}{\partial x} = -x & \lambda(1) &= \text{free} \\ 0 &= \frac{\partial H}{\partial u} = au + \lambda \Rightarrow u = -\frac{1}{a}\lambda\end{aligned}$$

- Differentiation of $\dot{x} = u = -\frac{1}{a}\lambda$ leads to

$$\ddot{x} = -\frac{1}{a}\dot{\lambda} = \frac{1}{a}x \Rightarrow x(t) = c_1 e^{\sqrt{1/a}t} + c_2 e^{-\sqrt{1/a}t}.$$

- With boundary conditions $x(0) = 1$ and $x(1) = 0$: $c_1 = \frac{1}{1 - e^{2\sqrt{1/a}}}$, $c_2 = \frac{-e^{2\sqrt{1/a}}}{1 - e^{2\sqrt{1/a}}}$

$\Rightarrow$ optimal solution:

$$\begin{aligned}x^*(t) &= \frac{1}{1 - e^{2\sqrt{1/a}}} \left(e^{\sqrt{1/a}t} - e^{\sqrt{1/a}(2-t)} \right), \\ u^*(t) &= \dot{x}^*(t) = \frac{\sqrt{1/a}}{1 - e^{2\sqrt{1/a}}} \left(e^{\sqrt{1/a}t} + e^{\sqrt{1/a}(2-t)} \right)\end{aligned}$$

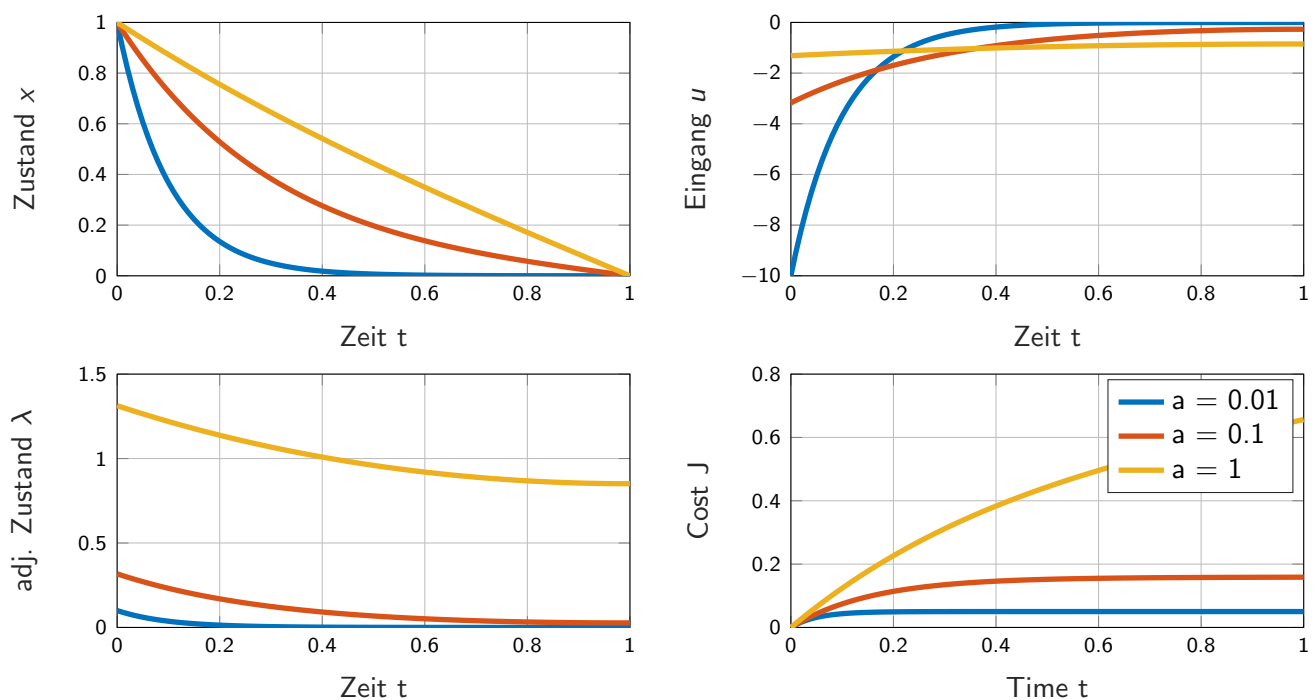


Figure 1: Trajectories for different parameter values.

- b) Higher value of a corresponds to higher weighting of u^2 in the cost function $\Rightarrow$ slower convergence of $x^*(t)$
 c) Interpretation of the adjoint variables: The adjoint variable $\lambda(0)$ at the initial time $t = 0$ represents a measure for the sensitivity of the cost functional regarding changes in the initial conditions $x(0)$.